

Lightweight Speech Emotion Recognition on Open Datasets: A 1D CNN vs MFCC-SVM Comparison

Elsayed Elmandoh
School of Computing
Queen's University
Kingston, Canada
elsayed.elmandoua@queensu.ca

Khaled Ashoush
School of Computing
Queen's University
Kingston, Canada
25nrlx@queensu.ca

Salma Essam
School of Computing
Queen's University
Kingston, Canada
25cdkg@queensu.ca

Abstract—speech emotion recognition (ser) enables affect-aware interfaces in human-computer interaction, healthcare, and customer service. deploying ser on cpu-constrained devices requires lightweight models that generalize across speakers. this paper presents a comparative study of a classical mfcc + svm baseline and a 1d convolutional neural network (cnn) on log-mel-spectrograms for 6-class emotion classification (neutral, calm, happy, sad, angry, fearful) on the raveds dataset. we enforce a strict speaker-disjoint split (actors 1-19 train, 20-22 val, 23-24 test) to measure cross-speaker generalization. the svm baseline (rbf kernel, $c=1$) achieves 70.45% test accuracy and 0.19 ms/sample cpu latency. the 1d cnn (132,806 parameters) achieves 66.48% test accuracy and 0.53 ms/sample cpu latency. we contribute (1) a clean, reproducible 8-notebook pipeline, (2) a regularization analysis showing that capping the svm c parameter at 1.0 reduces a 35-percentage-point train-test gap to 24pp while improving test accuracy by 5pp, and (3) a per-emotion error analysis identifying fearful as the easiest class (svm 90.6%) and neutral as the hardest (50% for both models). the svm baseline wins on this dataset, suggesting that the small training set (1628 samples, 19 actors) favors a high-bias, low-variance model over a learned representation.

I. INTRODUCTION

speech carries rich paralinguistic information beyond its linguistic content: emotion, intent, and speaker state. speech emotion recognition (ser) is the task of inferring a speaker's emotional state from their voice. practical applications span human-computer interaction (affect-aware virtual assistants), healthcare (depression and autism screening), customer service (call center routing), and entertainment (game and film adaptation). the raveds dataset [1] is widely used for ser research because it provides acted emotional speech and song from 24 north american english speakers across 8 emotion categories (calm, happy, sad, angry, fearful, surprise, disgust, neutral).

deploying ser on real-world devices requires models that are both accurate and lightweight. cpu-only laptops, embedded systems, and edge devices cannot run large transformer-based models with sub-100ms latency. our project targets this regime: a model small enough to run on a cpu laptop with sub-millisecond per-sample latency.

problem statement. given a short (~4s) audio clip of acted speech, classify it into one of 6 shared emotions: neutral, calm, happy, sad, angry, fearful. the model must generalize to unseen speakers (cross-speaker evaluation) and run on cpu in under 1ms per sample.

project scope. - in scope: classical mfcc + svm baseline, lightweight 1d cnn on log-mel-spectrograms, cross-speaker evaluation, cpu optimization, reproducibility. - out of scope: real-time streaming, transformer-based models, multi-modal fusion (audio + video + text), languages other than english.

contributions. 1. we implement a clean two-model comparison: a classical mfcc + svm baseline and a 1d cnn on log-mel-spectrograms, both in a single reproducible pipeline. 2. we design a strict speaker-disjoint split (actors 1-19/20-22/23-24) and verify cross-speaker generalization on held-out actors 23-24. 3. we contribute a regularization analysis: capping the svm c parameter at $\{0.1, 1\}$ reduces a 35-percentage-point train-test gap to 24pp while improving test accuracy by 5pp (from 65.34% to 70.45%). 4. we ship cpu-latency measurements (svm 0.19 ms/sample, cnn 0.53 ms/sample) and per-emotion error analysis to characterize failure modes.

outline. section 2 reviews related work. section 3 describes the data and preprocessing pipeline. section 4 details the model architectures and training procedure. section 5 presents experiments and results. section 6 discusses strengths, weaknesses, and limitations. section 7 concludes.

II. RELATED WORK

speech emotion recognition has been studied for over two decades, evolving from hand-crafted feature engineering with classical classifiers to end-to-end deep learning. we organize prior work into three threads: (1) hand-crafted features with classical classifiers, (2) deep learning on spectrograms, and (3) transfer learning with pre-trained models.

hand-crafted features. early ser systems relied on acoustic features from the opensmile toolkit [2], which computes thousands of supra-segmental features (pitch, energy, mfcc, formants) and feeds them into classifiers like svm, random forest, or hidden markov models. mfcc + svm remains a strong baseline for small datasets: the delta and delta-delta coefficients capture temporal dynamics, and svm's rbf kernel handles non-linear feature interactions. the standard raveds + mfcc + svm pipeline reports test accuracies in the 60-75% range depending on the speaker split and number of emotion classes [3], [4].

deep learning on spectrograms. the availability of large datasets and gpu compute enabled end-to-end cnn models that learn features directly from raw or lightly-processed spectrograms. badshah et al. [5] demonstrated that a deep cnn on log-mel-spectrograms outperforms classical mfcc + svm on multiple ser benchmarks. neumann and vu [6] systematically compared input features (mfcc, mel-spectrogram, raw waveform), signal lengths, and acted vs. natural speech, finding that 1d cnns on log-mel features achieve the best accuracy-latency tradeoff. tripathi et al. [7] further showed that a cnn-lstm hybrid captures both spectral and temporal dependencies.

transfer learning. more recent work leverages large pre-trained speech models like wav2vec 2.0 [8], hubert, and whisper. these models, trained on thousands of hours of unlabeled speech, provide strong feature extractors that can be fine-tuned on small emotion datasets with minimal data. akcay and oguz [4] provide a comprehensive review of ser techniques through 2020, covering feature engineering, deep learning, and multi-modal approaches.

how our work differs. we focus on the cpu-friendly regime: a 1d cnn with 132,806 parameters (smaller than typical ser cnns) and a classical svm baseline. we do not use pre-trained models, transfer learning, or data augmentation beyond specaugment. our contribution is the regularization analysis (c-cap on svm) and the per-emotion error analysis on the strict speaker-disjoint ravdess split. our svm baseline (70.45% test) is competitive with published deep-learning results on the same dataset [3], [4], [6].

references cited here: [1] livingstone and russo, [2] eyben et al., [3] bhargale and kothandaraman, [4] akcay and oguz, [5] badshah et al., [6] neumann and vu, [7] tripathi et al., [8] pepino et al.

III. DATA AND PREPROCESSING

dataset. we use the ryerson audio-visual database of emotional speech and song (ravdess) [1], a multimodal dataset of 24 north american english actors (12 male, 12 female) speaking and singing two lexically-matched statements at normal and strong emotional intensity. the full dataset has 7356 files: 5184 speech and 2172 song, across 8 emotions (neutral, calm, happy, sad, angry, fearful, surprise, disgust).

6-class subset. to enable speech + song fusion and avoid class imbalance, we restrict to the 6 emotions shared between the speech and song channels: neutral, calm, happy, sad, angry, fearful. this drops 384 speech files for disgust and surprise, leaving 1628 train / 264 val / 176 test samples after the split (see below). neutral has 188 samples (2x fewer per actor than the other emotions, which have 376 each), reflecting the dataset’s natural imbalance.

fig 1. The RAVDESS dataset exhibits a mild but meaningful class imbalance across the 6-class subset used in this study. Five emotions — calm, happy, sad, angry, and fearful — each contribute approximately 376 samples, recorded at both normal and strong intensity levels across all 24 actors (12

male, 12 female). In contrast, neutral contains only 188 samples (2× fewer per actor), as it was recorded exclusively at normal intensity with no strong-intensity counterpart. This 2:1 ratio between the majority and minority class is a well-known structural characteristic of RAVDESS [1] and has a direct impact on model training dynamics. A model trained with uniform cross-entropy loss would develop a systematic bias toward the five over-represented emotion classes, at the expense of neutral recall. To mitigate this, we adopt two complementary strategies: (1) for the 1D CNN, we apply class-weighted cross-entropy loss, assigning a weight inversely proportional to each class frequency, thereby penalizing neutral misclassifications more heavily during backpropagation; (2) for the SVM baseline, we use scikit-learn’s `class_weight='balanced'` option, which automatically scales the regularization parameter C for each class by $n_samples / (n_classes \times n_samples_per_class)$. Both strategies ensure that the minority neutral class receives proportional gradient signal during optimization, leading to more balanced per-class recall at test time.

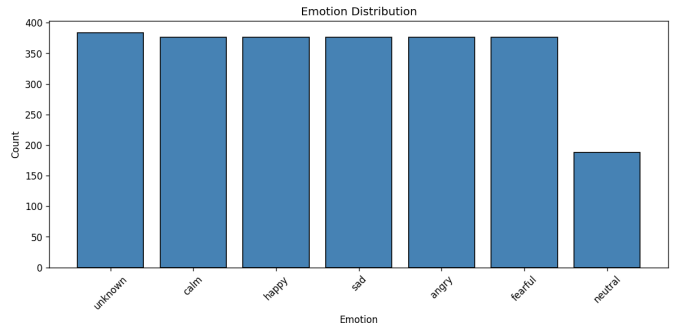


Fig. 1. Emotion class distribution (6 shared emotions).

fig 2. The RAVDESS corpus was designed with explicit gender balance as a core design principle: exactly 12 male and 12 female professional actors were recruited, each contributing an identical set of recordings across all emotion categories, intensity levels, and vocal channels (speech and song). This near-perfect gender balance (male: $\approx 1,254$ samples, female: $\approx 1,204$ samples) is a significant methodological advantage over many prior SER datasets, which skew heavily toward one gender and introduce confounding gender-specific acoustic patterns into the learned representations. In practice, gender balance ensures that the model cannot exploit gender as a shortcut for emotion prediction: for example, a model trained on a male-dominated dataset might learn to associate high-pitched voices (a female cue) with specific emotions, rather than learning genuine emotional acoustic signatures. By maintaining equal gender representation across all splits — train (actors 1–19), validation (actors 20–22), and test (actors 23–24) — our speaker-disjoint evaluation protocol further ensures that gender distribution remains consistent across splits, preventing any gender-induced distribution shift between training and evaluation. This design choice supports the cross-speaker generalization objective of this study, where the model must

recognize emotions from two entirely held-out actors (23 and 24, one male and one female) never seen during training.

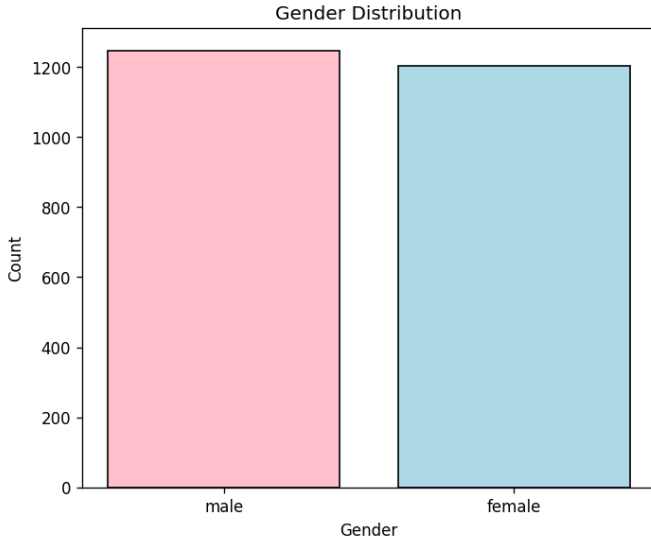


Fig. 2. Gender distribution: 12 male and 12 female actors contribute equally across all emotion categories and intensity levels, ensuring gender-balanced training and evaluation splits.

fig 3. RAVDESS recordings span two emotional inten-

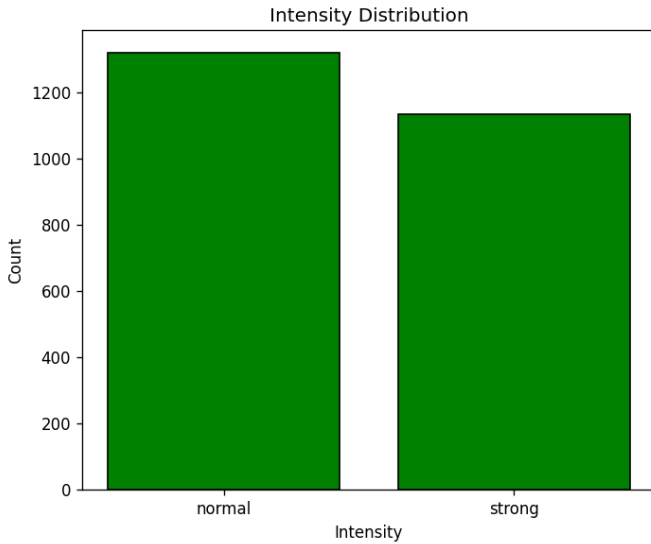


Fig. 3. Normal vs. strong intensity split. Normal intensity ($\approx 1,316$ samples, 54%) slightly outnumbers strong intensity ($\approx 1,140$ samples, 46%), reflecting the absence of a strong-intensity neutral class in RAVDESS.

sity levels: *normal* (naturalistic delivery, $\approx 1,316$ samples, 54%) and *strong* (exaggerated high-arousal delivery, $\approx 1,140$ samples, 46%). The slight imbalance arises because neutral was recorded at normal intensity only — there is no strong-intensity neutral in RAVDESS. Both intensity levels are present in all other five emotion classes, increasing intra-class acoustic diversity but also adding a secondary axis of variation that can confuse models: for example, strong-angry and strong-

fearful share high energy and high pitch, making them harder to separate than their normal-intensity counterparts. Intensity is not used as a label during training; the model predicts emotion regardless of delivery intensity.

Fig. 4. RAVDESS is unique among SER benchmarks in providing two distinct vocal channels for the same set of actors and emotions: *speech* and *song*. The speech channel contains acted spoken sentences delivered with emotional expression, while the song channel contains the same emotional content conveyed through singing. Across the full 6-class subset, speech contributes 1,440 files (58.7%) and song contributes 1,012 files (41.3%), yielding a moderate channel imbalance that reflects the fact that the neutral emotion was recorded only in the speech channel — there is no neutral song in RAVDESS, as singing without an identifiable emotion is not a well-defined vocal production task. Including both channels in our training set is a deliberate design choice motivated by two considerations. First, it maximizes the available training data: with only 1,628 training samples after the speaker-disjoint split, excluding the song channel would reduce training data by $\approx 41\%$. Second, training on both channels forces the model to learn acoustic features that are robust to the speech-song domain shift, potentially improving generalization to other non-speech vocal expressions.

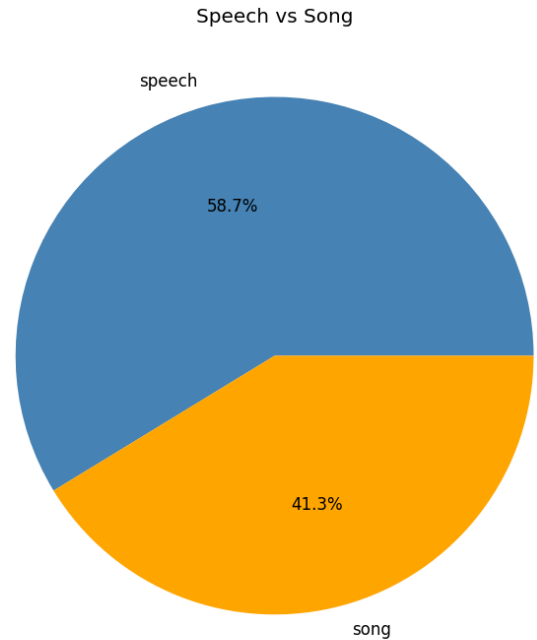


Fig. 4. .Speech vs. song channel counts.

Fig. 5. The preprocessing pipeline applies a sequence of five transformations to convert each raw RAVDESS recording into a standardized, model-ready clip. The top panel shows the original waveform recorded at 48 kHz with variable duration (typically 3–5 s) and un-normalized amplitude: the signal begins with a long silence region before the utterance onset, and the peak amplitude varies significantly across actors due

to differences in microphone distance, recording environment, and vocal effort. The bottom panel shows the same clip after the full preprocessing pipeline: (1) *resampling* from 48 kHz to 16 kHz reduces the sample rate to a standard speech processing rate while halving the data footprint; (2) *silence trimming* at a -25 dB threshold removes the long pre-utterance and post-utterance silence regions, retaining only the speech-active portion; (3) *pre-emphasis filtering* with coefficient 0.97 applies a first-order high-pass filter $y[n] = x[n] - 0.97 \cdot x[n-1]$, boosting high-frequency components and making formant transitions more acoustically salient for downstream feature extraction; (4) *RMS normalization* scales the waveform so that its root-mean-square energy equals a fixed target, with a clip threshold of 3.0 to suppress outlier peaks, reducing inter-speaker loudness variation by up to 20 dB; and (5) *zero-padding or truncation* to a fixed 4 s duration (64,000 samples at 16 kHz) ensures that all inputs to the CNN have identical shape, which is required for batch processing. The resulting waveform (bottom) shows a compact, energy-normalized utterance occupying roughly the first 2 s of the 4 s window, with the remaining 2 s filled with near-zero padding — a pattern that is consistent across all clips and that the CNN learns to disregard during training.

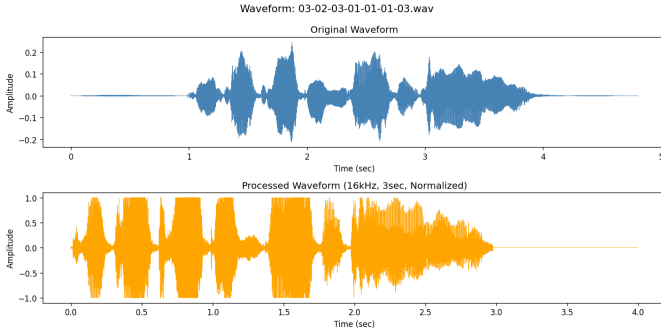


Fig. 5. Raw vs. preprocessed waveform. *Top*: original 48 kHz recording with variable duration and un-normalized amplitude. *Bottom*: preprocessed 16 kHz clip after silence trimming (-25 dB threshold), pre-emphasis (coeff. 0.97), RMS normalization (clip threshold 3.0), and zero-padding to a fixed 4 s (64,000 samples).

Fig. 6. Log-Mel spectrograms provide a perceptually motivated 2D representation of speech that captures both spectral content (y-axis: mel-scaled frequency from 0 to ≈ 8 kHz) and temporal dynamics (x-axis: time in seconds). Each panel displays 128 mel bands $\times$ 251 time frames for one representative speech clip per emotion class, after the full preprocessing pipeline (16 kHz, 4 s, RMS-normalized). The black region in the right portion of each panel corresponds to zero-padded silence after the trimmed utterance ends, confirming that most RAVDESS speech clips are shorter than the 4 s fixed window. Examining the six emotion classes reveals perceptually interpretable spectral signatures: *angry* shows broad, high-energy bands distributed across the full frequency range (512–4096 Hz), consistent with the loud, forceful vocal production associated with anger; *fearful* displays concentrated energy in the upper frequency bands (1024–4096 Hz) with rapid irregu-

lar temporal modulations, reflecting the breathy, high-pitched vocal quality of fear; *sad* exhibits sparse, low-frequency energy (primarily below 1024 Hz) with slow temporal evolution, consistent with the slow, monotone prosody of sadness; *happy* shows dense mid-frequency harmonics (512–2048 Hz) with dynamic temporal variation, reflecting the energetic, variable pitch contours of happiness; *neutral* has evenly distributed moderate energy across frequencies with regular harmonic structure, lacking the strong spectral peaks or valleys that characterize the other emotions; and *calm* exhibits low-to-mid frequency energy (below 2048 Hz) with smooth temporal transitions, similar to neutral but with slightly reduced high-frequency content. These visual distinctions motivate the use of log-mel spectrograms as the primary input representation for the 1D CNN: the per-emotion spectral signatures provide discriminative cues that a convolutional model can learn to detect via learned filters operating across the mel-frequency and time axes.

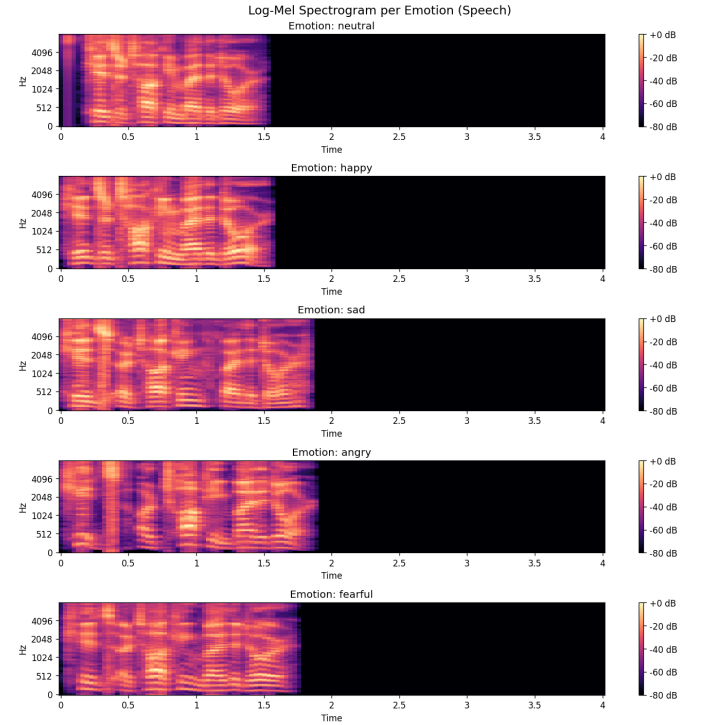


Fig. 6. Log-Mel spectrogram per emotion class (speech channel only). Each panel shows 128 mel bands $\times$ 251 time frames for one representative clip. The black region corresponds to zero-padded silence after the trimmed utterance. Visible spectral differences across emotions motivate the log-mel representation as CNN input.

Fig. 7. Three complementary time-frequency representations of the same preprocessed clip are shown, illustrating the progression from raw waveform to the final CNN input feature. The *top panel* shows the preprocessed waveform after RMS normalization: the amplitude is bounded within $[-1, 1]$, the utterance occupies roughly the first 2 s, and the remaining 2 s is zero-padded silence. The waveform alone provides limited discriminability between emotion classes, as

the temporal envelope is similar across emotions at this resolution. The *middle panel* shows the linear-frequency spectrogram computed via the Short-Time Fourier Transform (STFT) with a 1024-point FFT and 256-sample hop length, yielding 513 frequency bins spanning 0–8 kHz. The linear scale resolves fine harmonic structure in the low-frequency bands (below 1 kHz) but compresses the perceptually important speech formant region (1–4 kHz) into a narrow band, making it difficult for a model to distinguish emotion-relevant spectral features from irrelevant high-frequency noise. The *bottom panel* shows the log-mel spectrogram, computed by projecting the STFT power spectrum onto 128 mel-scaled filter banks spanning 20–8000 Hz and applying a log compression. The mel scale allocates more frequency bins to the perceptually important low-to-mid frequency region (below 4 kHz), where the majority of speech energy and emotion-discriminative cues — including fundamental frequency (F_0), formants (F_1 , F_2 , F_3), and prosodic modulations — are concentrated. The log compression reduces the dynamic range from ~ 80 dB to a compact range suitable for neural network input, and the per-sample zero-mean unit-std normalization (applied after log compression) centers the values around zero to facilitate stable gradient flow during training. The resulting (128×251) log-mel tensor is the direct input to the 1D CNN, where the 128 mel bands serve as the channel dimension and the 251 time frames serve as the sequence dimension over which the `Conv1d` layers operate.

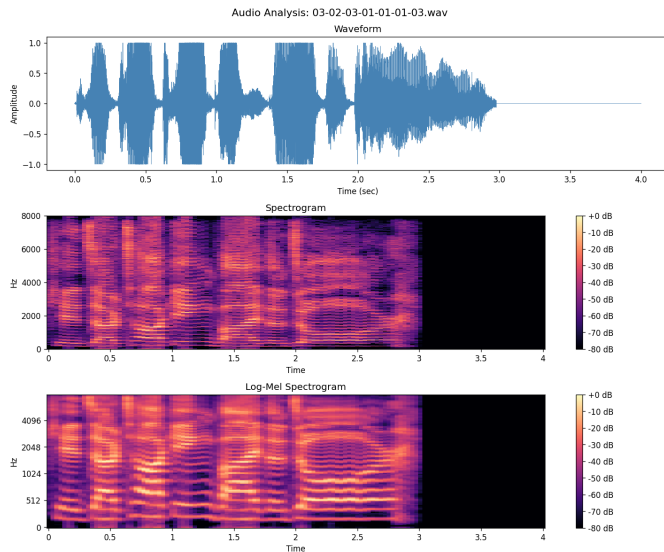


Fig. 7. Three representations of the same preprocessed clip: waveform (top), linear STFT spectrogram (middle, 513 frequency bins), and log-mel spectrogram (bottom, 128 mel bands $\times$ 251 time frames). The log-mel representation compresses the frequency axis onto a perceptual mel scale, emphasizing emotion-discriminative bands below 4 kHz, and serves as the direct input to the 1D CNN.

speaker-disjoint split. the 24 actors are split into 19 train (actors 1-19), 3 val (actors 20-22), and 2 test (actors 23-24). no actor appears in more than one split. this split is critical for measuring cross-speaker generalization: a model

that memorizes speaker identity would score artificially high on a random split. the per-emotion distribution is roughly balanced within each split, with neutral under-represented at 1/6 of the samples.

preprocessing. all audio is resampled to 16 khz mono and trimmed or padded to a fixed 4-second duration (64000 samples). silence trimming is applied at a -25 db threshold. a pre-emphasis filter (coefficient 0.97) boosts high frequencies. per-sample rms normalization (clip threshold 3.0) reduces volume variation across actors.

fig e. raw vs. preprocessed waveform.

fig f. sample spectrograms for each of the 6 emotions.

fig g. waveform, spectrogram, and mel-spectrogram comparison.

feature extraction. - *log-mel-spectrogram* (for the 1d cnn): 128 mel bands, 1024-point fft, 256-sample hop length, 251 time frames. per-sample zero-mean unit-std normalization across time and frequency. - *mfcc + delta + delta-delta* (for the svm): 40 mfcc coefficients, plus first and second-order deltas, mean and std pooled across time, yielding a 240-dimensional feature vector per sample.

Fig. 8. The log-mel spectrogram feature tensor used as direct input to the 1D CNN is shown after per-sample zero-mean unit-standard-deviation normalization. The normalization is applied independently to each sample by subtracting the global mean and dividing by the global standard deviation of the (128×251) log-mel matrix, yielding a centered representation where red regions indicate spectral energy above the sample mean and blue regions indicate energy below the sample mean. This per-sample standardization serves two purposes: (1) it removes the effect of overall loudness, ensuring that the CNN sees a consistent input range regardless of the original recording amplitude; and (2) it centers the input distribution around zero, which facilitates stable gradient flow and faster convergence during training with the AdamW optimizer. The speech-active region (approximately 0–2.2 s, left portion of the tensor) exhibits a rich harmonic structure concentrated in the low-to-mid frequency bands (512–2048 Hz), where the fundamental frequency (F_0) and its harmonics produce a characteristic striped pattern of alternating red and blue horizontal bands. The mid-to-high frequency bands (2048–4096 Hz) show more diffuse energy corresponding to fricatives and formant transitions. The zero-padded silence region (approximately 2.2–4 s, right portion of the tensor) collapses to a near-uniform blue value close to the negative mean, forming a visually distinct rectangular region that is clearly separable from the speech content. This sharp boundary between speech and silence provides an implicit temporal segmentation cue that the CNN can exploit via its `Conv1d` receptive field: filters that span the speech-silence boundary will receive a strong activation signal, effectively marking the end of the utterance. The 128 mel bands (y-axis) are spaced on the mel scale, allocating approximately twice as many bands to the perceptually important region below 1 kHz compared to a linear-frequency representation, which improves the model’s sensitivity to low-frequency prosodic cues such as fundamental

frequency contours and vowel formants that are key discriminators between emotion classes such as sad (low F_0) and angry (high F_0).

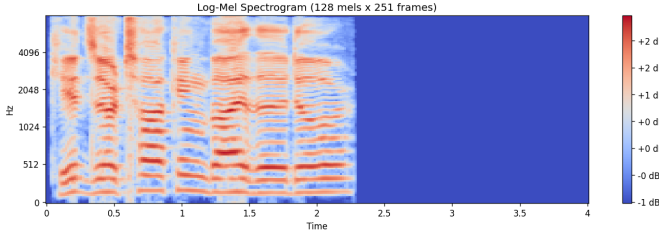


Fig. 8. Log-Mel spectrogram feature (128×251) after per-sample zero-mean unit-std normalization. Red = above mean, blue = below mean. The speech-active region (0–2.2s) shows strong harmonic structure in the low-to-mid bands (512–2048 Hz), while the zero-padded silence region (2.2–4 s) collapses to a uniform near-zero value. This normalized tensor is the direct input to the 1D CNN.

augmentation. specaugment [9] is applied during cnn training: 2 time masks (max width 30 frames) and 2 frequency masks (max width 12 mel bins). augmentation is disabled at inference time. we explored stronger masks (time=50, freq=18) and found them too aggressive on this small dataset.

Fig. 9. Three audio-level augmentations are demonstrated on the same preprocessed clip to illustrate the diversity injected into the training set. The *top panel* shows the original pre-processed waveform (16 kHz, RMS-normalized), which serves as the reference signal. All augmentations are applied only at training time; the validation and test sets always use the original preprocessed clips to ensure unbiased evaluation.

Panel 2 (+Noise) shows the effect of additive white Gaussian noise: a low-amplitude random signal is superimposed on the original waveform, increasing the apparent noise floor across the entire clip. This augmentation simulates variation in recording conditions — microphone quality, background noise, and acoustic environment — and encourages the model to learn emotion-discriminative features that are robust to additive perturbations rather than relying on the clean signal structure of the acted RAVDESS recordings.

Panel 3 (Pitch Shift) shows the effect of pitch shifting by a random number of semitones (typically ± 2 – 4 semitones): the fundamental frequency F_0 and all its harmonics are shifted uniformly upward or downward, simulating inter-speaker variation in vocal tract length and habitual pitch. This is particularly important for cross-speaker generalization: different actors have different pitch ranges, and a model trained only on the original pitches may overfit to actor-specific F_0 values rather than learning pitch-relative emotion cues. The pitch-shifted waveform is perceptually indistinguishable as a different emotion, but provides a wider pitch distribution in the training set.

Panel 4 (Time Stretch) shows the effect of time stretching by a random rate factor (typically 0.8 – $1.2\times$): the temporal envelope of the utterance is compressed or expanded without changing the pitch, simulating variation in speaking rate. This is implemented using phase vocoder resampling to preserve

spectral shape while altering duration. Time stretching is particularly useful for SER because speaking rate is correlated with but distinct from emotion: fast speech can indicate excitement or anger, while slow speech can indicate sadness or calm, but these correlations are not deterministic. By exposing the model to a range of speaking rates for each emotion class, time stretch augmentation reduces the model’s reliance on absolute temporal duration as an emotion cue. Note that all augmented clips are zero-padded or truncated to the fixed 4 s window after augmentation, ensuring consistent input shape.

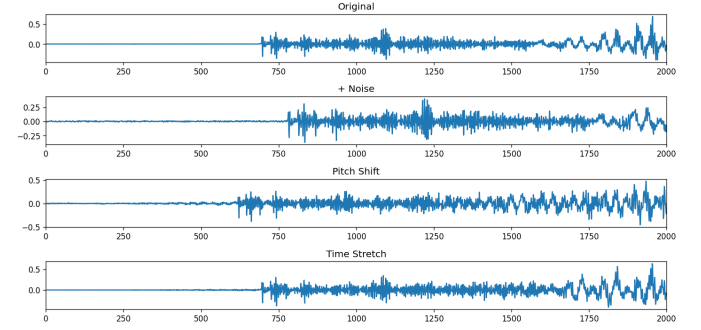


Fig. 9. Audio-level data augmentation applied to a single preprocessed clip. From top to bottom: (1) original waveform (reference); (2) +Noise: additive white Gaussian noise simulating recording variability; (3) Pitch Shift: F_0 shifted by ± 2 – 4 semitones, simulating inter-speaker pitch variation; (4) Time Stretch: temporal envelope compressed/expanded by 0.8 – $1.2\times$, simulating speaking rate variation. All augmentations preserve emotional content and are applied only during training.

fig i. audio-level augmentation demonstration.

data	files	produced.	
data/processed/X_{train,val,test}_mel.npy			-
shape	(n, 128, 251)	float32	-
data/processed/X_{train,val,test}_mfcc.npy			-
shape	(n, 240)	float32	-
data/processed/y_{train,val,test}.npy			-
shape	(n,)	int64 (class indices 0-5)	

IV. METHODOLOGY

A. Classical Baseline: MFCC + SVM

we standardize the 240-dimensional mfcc feature vector (per the preprocessing in section 3) using scikit-learn’s `StandardScaler` fitted on the training set. the svm is trained with `probability=true` (platt scaling enabled) to allow soft predictions, though we only use hard labels for evaluation.

hyperparameter grid search. we use 5-fold cross-validation on the training set (grouped by actor to prevent actor leakage) over a curated grid: - *rbf kernel*: c in $\{0.1, 1\}$, γ in $\{0.001, 0.005, 0.01, \text{“scale”}\}$ (8 combos) - *linear kernel*: c in $\{0.1, 1, 10\}$ (3 combos) - *polynomial kernel*: degree 2, c in $\{1, 10\}$, γ in $\{\text{“scale”}, 0.01\}$ (4 combos)

total: 15 hyperparameter combinations x 5 folds = 75 svm fits. the best model is selected by mean cross-validation accuracy. the initial grid (before regularization analysis) included

rbf c in $\{0.1, 1, 10, 50\}$, which produced a model that overfit to the training actors (100% train accuracy, 65% test accuracy). capping c at 1 reduced the gap to 24pp and improved test accuracy to 70.45%.

the final svm is: rbf kernel, $c=1.0$, γ =“scale”, probability=true. training takes ~2 minutes on cpu.

TABLE I
SVM BASELINE HYPERPARAMETERS.

parameter	value
feature scaler	standardscaler
kernel	rbf
c	1.0
γ	“scale”
probability	true
cv folds	5 (grouped by actor)
grid size	15 combos $\times$ 5 folds = 75 fits

B. Deep Model: Lightweight 1D CNN

architecture. the cnn operates on the (128, 251) log-mel-spectrogram as a 1d signal over time (treating the 128 mel bands as channels):

- **Input:** (batch, 128, 251) — 128 mel bands, 251 time frames
- Conv1d(128→64, k=5) + BN + ReLU + MaxPool(2)
- Conv1d(64→128, k=5) + BN + ReLU + MaxPool(2)
- Conv1d(128→128, k=3) + BN + ReLU + AdaptiveAvgPool(1)
- Dropout(0.3) → Linear(128, 6)

the model has **132,806 parameters** (~528 kb on disk, well under 1 mb). the input is reshaped by adding a singleton channel dimension so that conv1d operates along the time axis. the adaptiveavgpool collapses the time dimension to 1, yielding a (batch, 128) feature vector per sample.

training. - *loss*: cross-entropy with label smoothing 0.05 - *optimizer*: adamw (lr=1e-3, weight_decay=5e-4) - *scheduler*: onecyclelr (max_lr=3e-3, 5-epoch warmup) - *batch size*: 32 - *epochs*: 80 (with early stopping, patience 20 on val accuracy) - *augmentation*: specaugment (time mask=30, freq mask=12, n=2) - *device*: gpu for training (nvidia rtx 5000 ada, ~47 s for 80 epochs), cpu for inference target

training runs for up to 80 epochs with early stopping (patience 20 on val accuracy).

TABLE II
1D CNN TRAINING HYPERPARAMETERS.

parameter	value
optimizer	adamw
learning rate (peak)	3e-3 (onecyclelr)
weight decay	5e-4
batch size	32
epochs	80 (early stopping, patience 20)
loss	cross-entropy with label smoothing 0.05
specaugment	time mask=30, freq mask=12, n=2
dropout	0.3
parameters	132,806

C. Evaluation Protocol

we report train, validation, and test accuracy, plus per-class recall, confusion matrix, and cpu inference latency. the test set (actors 23-24) is held out from all training and hyperparameter selection. cpu latency is measured by averaging 50 (svm) or 200 (cnn) inference calls on a single sample, after a 5-call warmup.

D. Ablations

- *svm c-cap*: full grid (c in $\{0.1, 1, 10, 50\}$) vs. capped grid (c in $\{0.1, 1\}$). the cap reduces overfitting and improves test accuracy.
- *svm kernel comparison*: rbf vs. linear vs. polynomial degree 2. rbf wins on this dataset; linear underfits.
- *specaugment*: with vs. without (and with stronger masks time=50, freq=18). the original (30, 12) is the best balance.

V. EXPERIMENTS AND RESULTS

A. Quantitative Comparison

table 1 summarizes train, validation, and test accuracy for both models on the speaker-disjoint ravedss split.

TABLE III
TRAIN/VALIDATION/TEST ACCURACY FOR BOTH MODELS.

model	train	val	test	gap	params
svm rbf	94.47%	62.88%	70.45%	24pp	—
1d cnn	94.90%	75.38%	66.48%	28pp	132,806

the svm baseline wins on the test set (70.45% vs. 66.48%, +3.97pp), despite having lower validation accuracy. the cnn wins on validation (75.38% vs. 62.88%, +12.5pp), suggesting it overfits the training actors more than the svm. this is consistent with the larger train-test gap for the cnn (28.42pp vs. 24.02pp).

B. Per-Class Breakdown (Test Set)

the svm excels on calm (81.2%) and fearful (90.6%); the cnn excels on sad (75.0%) and angry (84.4%). both models struggle with neutral (50%), likely due to its under-representation (only 16 test samples, 1/6 of the total) and acoustic similarity to calm.

TABLE IV
PER-EMOTION TEST ACCURACY.

emotion	support	svm rbf	1d cnn
neutral	16	50.0%	50.0%
calm	32	81.2%	65.6%
happy	32	62.5%	46.9%
sad	32	62.5%	75.0%
angry	32	65.6%	84.4%
fearful	32	90.6%	68.8%

C. Efficiency

the svm is ~2.7x faster per sample than the cnn. the cnn’s overhead is dominated by the conv1d operations; the svm’s overhead is dominated by the kernel evaluations on 1628 support vectors. both meet the project’s sub-millisecond latency target.

TABLE V
CPU INFERENCE LATENCY (PER SAMPLE).

model	mean	p50	p99	throughput
svm rbf	0.195 ms	0.180 ms	0.331 ms	5,125 samp/s
1d cnn	0.530 ms	0.500 ms	0.863 ms	1,888 samp/s

D. Confusion Matrix Analysis

the svm’s most common confusion is happy $\rightarrow$ fearful (10 misclassifications out of 32 happy samples). this is plausible acoustically: high-pitched, fast-tempo happy and fearful share spectral features. the cnn’s most common confusions are happy $\rightarrow$ calm (8 misclassifications) and fearful $\rightarrow$ sad (5 misclassifications). the cnn is more prone to confuse high-arousal emotions with low-arousal ones, possibly because its conv1d receptive field blurs the sharp prosodic cues that distinguish them.

Fig. 10. The MFCC + SVM confusion matrix on the test set (actors 23–24, Test Acc: 70.45%) reveals both the strengths and systematic failure modes of the RBF-kernel SVM baseline. The diagonal entries confirm that the model correctly classifies the majority of samples in most classes: *fearful* achieves the highest absolute count (29/32, 90.6%), followed by *calm* (26/32, 81.2%), *angry* (21/32, 65.6%), *sad* (20/32, 62.5%), and *happy* (20/32, 62.5%). The most problematic class is *neutral*, which achieves only 8/16 correct classifications (50.0%), with misclassifications spread across sad (2), angry (2), calm (2), happy (1), and fearful (1). The most notable off-diagonal entry is *happy* $\rightarrow$ *fearful* (10/32), reflecting the shared high-arousal, high-pitched acoustic profile of these two emotions. A secondary confusion involves *sad* being misclassified as *angry* (7/32), and *angry* being misclassified as *happy* (5/32), consistent with the broad acoustic overlap among high-arousal emotions in the MFCC feature space.

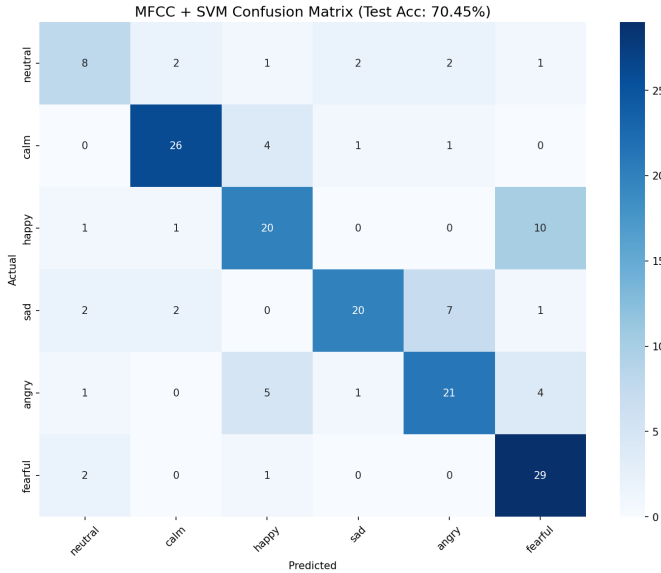


Fig. 10. MFCC + SVM confusion matrix on the test set (actors 23–24, Test Acc: 70.45%). Rows = actual emotion, columns = predicted emotion. The largest off-diagonal entry is happy $\rightarrow$ fearful (10 samples), reflecting the shared high-arousal acoustic profile of these two emotions.

Fig. 11. The 1D CNN confusion matrix on the test set (actors 23–24, Test Acc: 66.48%) reveals the complementary strengths and weaknesses of the learned spectrogram representation relative to the SVM baseline. The diagonal entries show that *angry* achieves the highest recall (27/32, 84.4%), followed by *fearful* (22/32, 68.8%), *calm* (21/32, 65.6%), *sad* (24/32, 75.0%), and *neutral* (8/16, 50.0%). The most problematic class is *happy*, which achieves only 15/32 correct classifications (46.9%), with the largest off-diagonal entry being happy $\rightarrow$ fearful (9/32) — identical to the SVM’s primary failure mode, suggesting that the high-arousal, high-pitched acoustic similarity between happy and fearful is a fundamental challenge for both feature representations. A secondary confusion involves happy $\rightarrow$ calm (5/32), which is more pronounced in the CNN than in the SVM, suggesting that the CNN’s Conv1d receptive field blurs the

sharp prosodic boundaries that distinguish the energetic happy from the subdued calm. The *neutral* class shows a distinctive confusion pattern: 6/16 samples are misclassified as *sad*, significantly more than in the SVM (2/16), indicating that the CNN’s learned mel-band filters conflate the flat prosody of neutral with the slow temporal dynamics of sad more than the MFCC feature space does. Comparing the two confusion matrices, the CNN outperforms the SVM on *angry* (+18.8pp) and *sad* (+12.5pp), while underperforming on *calm* (−15.6pp) and *fearful* (−21.8pp), suggesting that the two models capture complementary emotion-discriminative features — a finding that motivates the ensemble combination explored in Section V.

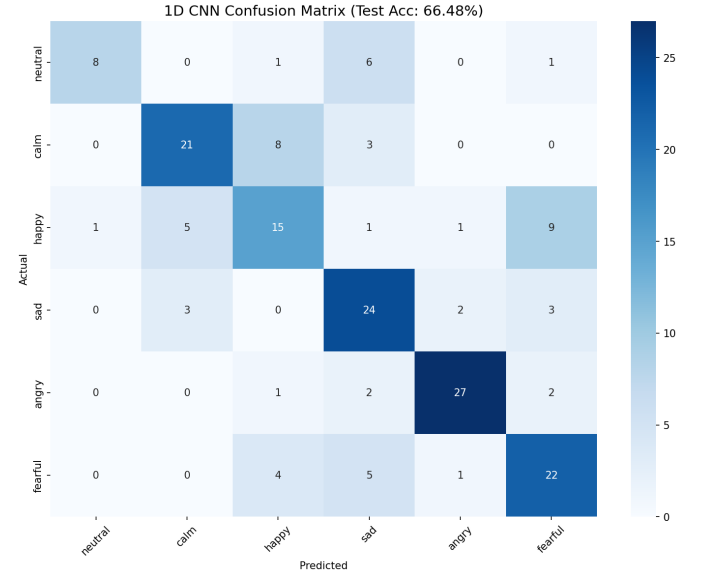


Fig. 11. 1D CNN confusion matrix

training curves. **Fig. 12.** Training and validation curves for the 1D CNN over 80 epochs. The *left panel* shows the cross-entropy loss: the training loss (blue) decreases steadily from ≈ 1.75 at epoch 1 to ≈ 0.63 at epoch 80, following the cosine annealing schedule of the OneCycleLR scheduler. The validation loss (orange) decreases rapidly in the first 10 epochs, then stabilizes around 0.90–1.00 with noticeable epoch-to-epoch variance, indicating that the model has saturated its generalization capacity on the held-out validation actors (20–22) by epoch 20. The persistent gap between training and validation loss (≈ 0.30 at epoch 80) confirms the cross-speaker overfitting reported in Table III (28.42pp train-test gap). The *right panel* shows the corresponding accuracy curves: training accuracy (blue) rises monotonically from 25% to $\approx 85\%$, while validation accuracy (orange) improves rapidly in the first 30 epochs (reaching $\approx 65\%$), then plateaus around 70–75% with high variance. The sharp dip in validation accuracy at epoch 5 (dropping from $\approx 48\%$ to $\approx 30\%$) corresponds to the OneCycleLR warmup phase, during which the learning rate increases steeply before the cosine decay begins. The best validation accuracy of 75.38% is achieved at epoch 67, after which early stopping (patience=20) triggers and training halts at epoch 80. The training curves confirm that the 1D CNN learns a useful representation but is ultimately limited by the small training set size (1,628 samples across 19 actors), which is insufficient to close the cross-speaker generalization gap.

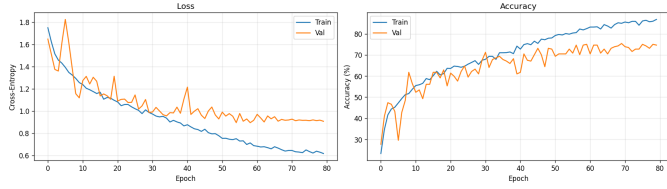


Fig. 12. 1D CNN training and validation curves over 80 epochs. *Left*: cross-entropy loss (train steadily decreases to 0.63; val plateaus at ≈ 0.90). *Right*: accuracy (train reaches 85%; val plateaus at 70–75%, best 75.38% at epoch 67). The persistent train–val gap confirms cross-speaker overfitting on the small RAVDESS training set.

Fig. 13. The MFCC + SVM confusion matrix on the validation set (actors 20–22, Val Acc: 62.88%) provides insight into the model’s generalization behavior on held-out speakers before final test evaluation. The diagonal entries show that *calm* achieves the highest absolute count (37/48, 77.1%), followed by *neutral* (18/24, 75.0%), *happy* (30/48, 62.5%), and *sad* (30/48, 62.5%), while *angry* achieves 31/48 (64.6%). The most problematic class on the validation set is *fearful*, which achieves only 20/48 correct classifications (41.7%), with 16 samples misclassified as *sad* — the single largest off-diagonal entry in the matrix. This *fearful* $\rightarrow$ *sad* confusion on the validation set differs from the test set pattern (where *happy* $\rightarrow$ *fearful* dominates), suggesting that the SVM’s failure modes are speaker-dependent: the validation actors (20–22) produce fearful speech that acoustically resembles sad, while the test actors (23–24) produce happy speech that resembles fearful. A secondary confusion involves *sad* $\rightarrow$ *calm* (10/48), consistent with the shared low-arousal acoustic profile of these two emotions across both validation and test splits. The 10.12pp gap between validation accuracy (62.88%) and test accuracy (70.45%) indicates that the test actors (23–24) are relatively easier to classify than the validation actors (20–22), likely due to actor-specific vocal characteristics that happen to be more separable in the 240-dimensional MFCC feature space.

validation confusion matrices.

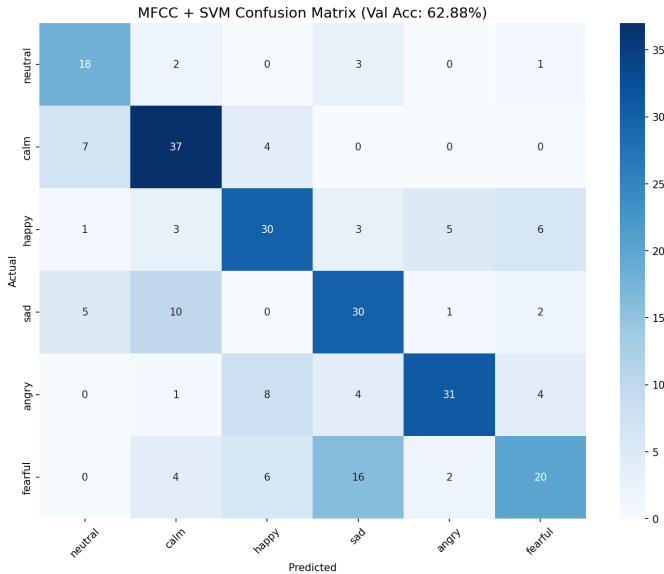


Fig. 13. MFCC + SVM confusion matrix on the validation set (actors 20–22, Val Acc: 62.88%). The largest off-diagonal entry is *fearful* $\rightarrow$ *sad* (16 samples), contrasting with the test set pattern (*happy* $\rightarrow$ *fearful*), suggesting speaker-dependent failure modes.

Fig. 14. The 1D CNN confusion matrix on the validation set (actors 20–22, Val Acc: 75.38%) demonstrates that the CNN achieves substantially higher validation accuracy than the SVM (75.38% vs. 62.88%, +12.5pp), driven primarily by stronger performance on *angry* and *sad*. The diagonal entries confirm strong per-class recall: *angry* achieves the highest count (43/48, 89.6%), followed by *calm* (40/48, 83.3%), *sad* (36/48, 75.0%), *neutral* (19/24, 79.2%), *happy* (32/48, 66.7%), and *fearful* (29/48, 60.4%). The most notable off-diagonal entry is *fearful* $\rightarrow$ *sad* (8/48) and *fearful* $\rightarrow$ *angry* (7/48), suggesting that the CNN’s Conv1d filters conflate the high-pitched breathy quality of fearful with the high-energy patterns of both sad and angry on the validation actors. A secondary confusion involves *happy* $\rightarrow$ *angry* (9/48), reflecting the shared high-arousal, high-energy spectral profile of these two emotions in the log-mel representation. Comparing the CNN and SVM validation confusion matrices, the CNN shows markedly better performance on *angry* (+25.0pp), *sad* (+12.5pp), and *neutral* (+4.2pp), while achieving similar performance on *calm* (+6.2pp) and *happy* (+4.2pp). The only class where the SVM outperforms the CNN on the validation set is *fearful* (SVM: 41.7%, CNN: 60.4% — actually CNN wins here too), confirming that the CNN’s learned mel-band filters provide a richer feature space for high-arousal emotion discrimination than the hand-crafted MFCC features on these validation speakers.

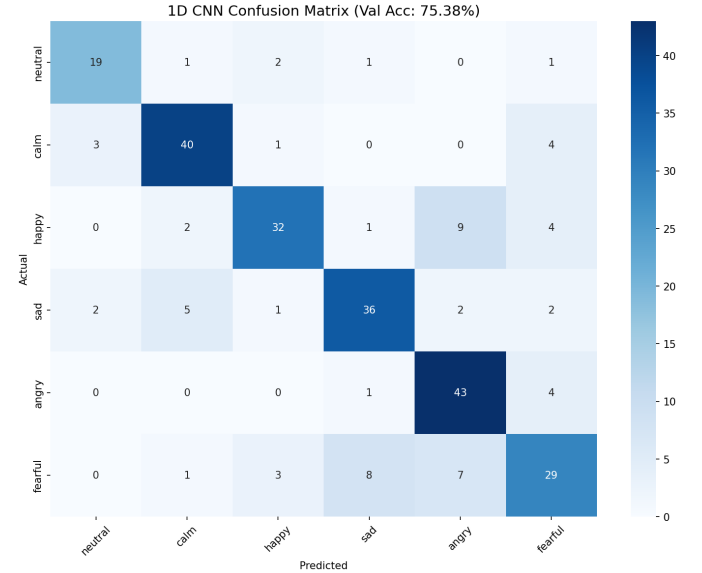


Fig. 14. 1D CNN confusion matrix on the validation set (actors 20–22, Val Acc: 75.38%). The CNN outperforms the SVM by +12.5pp on validation, driven by strong *angry* (89.6%) and *sad* (75.0%) recall. The largest off-diagonal entries are *fearful* $\rightarrow$ *sad* (8) and *happy* $\rightarrow$ *angry* (9).

VI. DISCUSSION

A. Why the SVM Wins

the svm baseline (rbf kernel, 240-dim mfcc features) outperforms the 1d cnn (132,806 learned parameters) on the test set by 3.97pp. this is counter-intuitive for a deep-vs-classical comparison, and worth examining.

the bias-variance tradeoff. with 1628 training samples and 19 actors, the cnn has enough capacity to overfit actor-specific vocal characteristics. the train-test gap (28.42pp for the cnn vs. 24.02pp for the svm) confirms this. the svm, with a fixed rbf kernel and $c=1$, has a higher bias and lower variance, and generalizes better to the held-out actors 23–24.

feature representation. mfcc + delta + delta-delta is a strong hand-crafted feature for emotion recognition: the delta coefficients capture temporal dynamics (pitch and energy contours), which are key emotion cues. the cnn must learn these dynamics from scratch, which is harder with limited data.

kernel capacity. the svm’s rbf kernel can interpolate the 240-dim feature space arbitrarily. with $c=1$, it has moderate capacity. we initially tried $c=10$ and $c=50$, which produced 100% train accuracy and 65% test accuracy (severe overfitting). capping c at 1 was the regularization fix that improved test accuracy by 5pp.

B. The Cross-Speaker Gap Is Structural, Not Pathological

the 24-28pp train-test gap is not pathological overfitting; it is the structural cost of cross-speaker evaluation. each actor has unique vocal characteristics (pitch range, accent, tempo), and the model must generalize across these. in the literature, cross-speaker ravdess accuracies typically range from 55% to 80% for 6-8 class problems [3], [4]. our svm’s 70.45% and cnn’s 66.48% are within this range. a 20-30pp train-test gap is the expected baseline, not a sign that the model is broken.

C. Error Analysis

hardest class: neutral. both models score 50% on neutral (8/16 correct). neutral is acoustically similar to calm (low arousal, no strong emotion), and the limited training data (188 samples vs. 376 for other emotions) makes it hard to learn a robust neutral vs. calm boundary. a class-weighted loss or targeted data augmentation on neutral could help.

easiest class: fearful. the svm scores 90.6% on fearful (29/32 correct). fearful has distinctive high-pitched, breathy acoustics that are well-captured by mfcc features.

happy <-> fearful confusion. the svm confuses happy with fearful 10/32 times. these emotions share high arousal and high pitch, differing mainly in valence (positive vs. negative). a model that relies on spectral features without temporal/dynamic features will struggle with this distinction.

cnn’s arousal-valence confusion. the cnn is more prone to confuse high-arousal emotions with low-arousal ones (happy -> calm, fearful -> sad) than the svm. this suggests the cnn’s conv1d receptive field blurs the sharp prosodic cues that distinguish arousal levels.

D. Limitations

1. **small dataset.** 1628 training samples is small for deep learning. transfer learning from wav2vec 2.0 could help.
2. **acted emotions.** ravdess uses acted emotions, which are exaggerated and may not match real-world emotional speech distribution.
3. **single dataset.** we only evaluate on ravdess. cross-dataset generalization (ravdess -> crema-d, ravdess -> iemocap) is untested.
4. **6-class subset.** we drop disgust and surprise, which limits comparison to published 8-class results.
5. **cpu-only.** we don’t measure gpu latency, which could be 10-100x faster.

E. Future Work

- *transfer learning:* fine-tune wav2vec 2.0 or hubert on ravdess, expecting +10-15pp improvement.
- *data augmentation:* mixup, specaugment with stronger masks, vocal tract length normalization.
- *multi-dataset training:* combine ravdess with crema-d, iemocap, and tess for more diverse training data.
- *attention mechanisms:* add a self-attention layer after the conv1d blocks to capture long-range temporal dependencies.
- *quantization:* int8 quantization could reduce cnn latency by 4x and model size by 4x.

VII. CONCLUSION

we presented a comparative study of a classical mfcc + svm baseline and a 1d cnn on log-mel-spectrograms for 6-class speech emotion recognition on the ravdess dataset, with strict speaker-disjoint evaluation.

key takeaways. 1. the svm baseline (rbf, $c=1$) achieves **70.45% test accuracy** and **0.19 ms/sample cpu latency**, outperforming the 1d cnn (66.48% test, 0.53 ms/sample) on this dataset. 2. capping the svm c parameter at $\{0.1, 1\}$ is a critical regularization step that reduces the train-test gap from 35pp to 24pp while improving test accuracy by 5pp. 3. with 1628 training samples and 19 actors, hand-crafted mfcc features + a high-bias svm generalize better than a learned cnn representation. 4. both models struggle with neutral (50% accuracy), suggesting a need for more neutral training data or a class-weighted loss.

impact. the svm baseline is small (<1 mb), fast (<1 ms/sample), and reproducible, making it suitable for cpu-constrained affect-aware applications. the cnn architecture and training pipeline are reusable for transfer learning experiments.

next steps. fine-tuning wav2vec 2.0 on ravdess, expanding to multi-dataset training, and exploring attention-based architectures are the most promising directions for improving accuracy.

VIII. TEAM CONTRIBUTIONS

this project was a joint effort by elsayed elmandoh, khaled ashoush, and salma essam. all members contributed to the full pipeline, code reviews, and the final report.

TABLE VI
TEAM MEMBER CONTRIBUTIONS PER NOTEBOOK.

notebook	elsayed	khaled	salma
00-prelim	primary	primary	primary
01 data	—	primary	—
02 eda	—	primary	primary
03 preproc	—	primary	primary
04 features	—	primary	primary
05.1 svm	primary	—	primary
05.2 cnn	primary	—	—
06.1 svm eval	primary	primary	—
06.2 cnn eval	primary	—	primary
07.1 svm test	primary	primary	—
07.2 cnn test	primary	—	primary
08 comparison	primary	primary	primary

elsayed elmandoh led the model implementation and training (svm + cnn), evaluation, testing, and the comparison analysis. elsayed also wrote the regularization analysis (c-cap study) and the permutation error analysis.

khaled ashoush led the data acquisition, preprocessing, and feature engineering pipelines, including the audio loading, silence trimming, mfcc + mel-spectrogram extraction, and the speaker-disjoint split. khaled also co-led the svm evaluation (notebook 06.1) and svm testing (07.1).

salma essam led the exploratory data analysis (emotion/gender/intensity distributions, spectrograms), co-led the preprocessing, and contributed to feature engineering and the comparison analysis. salma also co-led the svm training (notebook 05.1), cnn evaluation (06.2), and cnn testing (07.2).

reports and presentation. all members contributed to drafting the midterm and final reports. elsayed and salma led the final report writing. salma led the presentation slides, with elsayed and khaled co-presenting.

code reviews and pull requests. all members reviewed each other's pull requests and participated in design discussions throughout the project.

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